Rank-27 Reserved Assignment-Trail Representation

Tactix Development Project

24 August 2026

Executive summary

Current-Tactix/current-Kissat full-run profiling localized a portable representation cost in Tactix's propagation path: every assignment called the generic grow-capable `std::vector<unsigned>::push_back`, although variable growth had already reserved enough trail space and search can assign at most one literal per variable. Kissat instead writes through a reserved raw-stack cursor. The general repair introduces a Kernel-owned pre-sized backing store with a separate logical end. It changes no proof rule, strategy admission, schedule, logical effort currency, candidate order, or benchmark-specific arm.

At the exact conflict-9450 common state, the repair preserves every non-time counter and the complete event trace while reducing sampled cycles by 4.31% and total time by 748 ms. The mandatory frozen Tier B plus fixed Tier C guard passes 66/66, and exact-source ASan/UBSan native-LRAT fuzz passes 10,000 cases with zero failures. The checked immutable Rank-27 witness preserves every non-time statistic and the exact 1,514,395,111-byte proof while improving 413.830 to 407.006 internal seconds. Pinned `drat-trim` verifies that proof, and all campaigns leave zero certificate residue. This is a validated 6.824-second same-trajectory recovery, but Rank 27 remains open at 1.357 times the 300-second engineering target.

1 Findings and results

Current/current source localization

The immutable input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Frozen Kissat 4.0.4 proves UNSAT in 278.187 seconds and was not rerun. Current instrumented Kissat is diagnostic-only, never scoring evidence.

Full paired cycles campaign `benchmarks/runs/20260824-1405-tierd-rank27-full-paired-profile/`, job `1645700`, exactly reproduces both accepted trajectories. Tactix returns UNSAT in 436.73 CPU / 440.95 profiled wall seconds with 3,813,669 conflicts, 175,272,633 decisions, and 2,512,356,477 propagations. Current-source Kissat returns UNSAT in 288.17 / 290.82 seconds and reproduces its accepted diagnostic elimination counters. Both profiles have zero lost samples. Watched propagation leads both at 32.66% and 42.47% of samples; normalizing sampled cycles by printed propagations gives approximately 215.5 versus 174.2 cycles per propagation. That normalization is a localization lead, not proof of equal full-run work because the trajectories differ.

Source and disassembly comparison then isolates the first narrow difference. Tactix's already-reserved assignment trail still invokes an out-of-line shared vector push owner, accounting for another 1.90% of full-run samples. Kissat's equivalent hot path directly stores through its reserved stack cursor. No candidate, schedule, or deduction differs at that representation boundary.

General representation repair

`ReservedTrail` owns a pre-sized `std::vector<Lit>` plus a logical `used` end. `reserve` enlarges backing size only on variable growth; `push_back` asserts spare storage and performs one indexed store; backtrack changes only the logical end. Variable compaction adopts the replacement vector allocation and exposes its spare capacity as backing size before search resumes. Iteration, indexing, conversion, and Kernel clients see only the live prefix. A focused regression covers reserve, push, iteration, conversion, shrink, pop, compaction-vector adoption, and post-adoption push.

This contract follows an existing semantic invariant—at most one assignment per variable, with reservation before any new variable can be assigned. It does not introduce a capacity constant or a formula classifier. The exact Release candidate SHA-256 is `b6c1623b89477b140cabddb3cb139ca797778a871aeb67ab35fc708c3cd87851`; its source diff SHA-256 is `213a9e4ce96b9f4af2e72dad0f571864c31a8b3823bf85dd8ebb9965a7897d6c`.

2 Evidence

Exact common state and trace

Campaign `benchmarks/runs/20260824-1410-tierd-rank27-reserved-trail-profile/`, job `1645702`, runs predecessor and candidate simultaneously to conflict 9450 under the same two-CPU/16-GiB contract. Both have exactly 9,450 conflicts, 5,795,248 decisions, 96,338,812 propagations, 16 Vivify rounds, and identical printed rule work, logical ticks/costs, scheduling, and pruning. Candidate time is 16.624 seconds total / 8.090 inprocessing / 8.533 search and scaffolding, versus 17.372 / 8.301 / 9.071 for the predecessor. Sampled cycles fall 66,658,958,306 to 63,787,396,653, or 0.9569 times; the shared vector-push owner falls 2.78% to 0.63%, with no assignment-propagator push/emplace call left. Both profiles have zero lost samples and zero residue.

Event campaign `benchmarks/runs/20260824-1415-tierd-rank27-reserved-trail-trace/`, job `1645704`, emits 64 untruncated rows byte-identical to the accepted predecessor, SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`. This freezes exact rule boundaries, candidate identities and order, normalized ticks and costs, scheduling order, resulting learning, and pruning through the conflict-9449 Probe and conflict-9450 stop. It also preserves the completed Sweep parity, counted directed bridge repair, BVE antecedent admission, and proof-rule/strategy boundary.

Mandatory guard and fuzz

Full-width campaign `benchmarks/runs/20260824-1420-BC-reserved-trail-guard/`, array `1645705` plus aggregate `1645771`, runs all frozen 28 Tier-B inputs and the predeclared five mechanism-relevant Tier-C inputs under 16 GiB, 1,200 seconds, and two CPUs. Both candidate and shipped arms validate 17 SAT models and 16 UNSAT proofs: 66/66 terminal and valid, with zero invalid, missing, regression, cleanup, or residue entry. Candidate PAR-2 is 1,680.217 seconds versus 2,218.414 shipped, ratio 0.757395599. Summary SHA-256 is `c2ceadd24063d7b067f1d3872b7dd124abd61da6e9175f8d895645114b5a7909`.

Exact-source sanitizer campaign `benchmarks/runs/20260824-1425-reserved-trail-fuzz/`, job `1645772`, completes 10,000 cases with zero failures: 5,188 independently checked SAT models, 3,577 checked native-LRAT UNSAT proofs, and 1,235 declared bounded nonterminations. BVE, Substitute, and Sweep are reached in 6,163, 5,141, and 2,755 cases, totaling 1,459,585 target applications. There are zero sanitizer findings, rejected models, rejected proofs, or crashes. Twenty-five timeout CNFs remain as diagnostic inputs only; proof/certificate residue is zero. Random

generation did not reach VariableCompaction, so the focused ASan/UBSan Kernel regression is the direct compaction/adoption authority. No soundness trigger fired and the bounded Claude repair workflow was not invoked.

Checked immutable Rank 27

Campaign `benchmarks/runs/20260824-1445-tierd-rank27-reserved-trail-witness/`, job `1645778`, returns expected UNSAT in 407.006 internal / 407.497 wall seconds at 665,356 KiB solver RSS. The preceding checked pin needed 413.830 / 414.375 seconds, so the repair recovers 6.824 internal and 6.878 wall seconds on an identical trajectory. All conflicts, decisions, propagations, rule yields, and other non-time statistics match exactly. Inprocessing falls 167.753 to 166.746 seconds and search/scaffolding 246.078 to 240.260; individual timers are noisy but Vivify and BVE fall 55.877 to 54.854 and 31.122 to 30.466.

The proof is byte-identical: 1,514,395,111 bytes, SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`. Pinned `drat-trim` verifies it in 112.088 wall seconds using 20,047,408 resolution steps and zero RAT lemmas in the core. Proof, decompressed input, checker derivatives, and scratch are absent before result publication. Result SHA-256 is `2c16b3697c5bdec2afae16cd8cc1de394429723d8e2ba45d7ca5b19e771ce2c7`.

The cumulative Release suite is 73/74: executable, proof/model, configuration, trace, static boundary, and cleanup tests pass; only the already recorded opt-in documentation-size debt fails. Focused ASan/UBSan Kernel coverage is green. Repository authority forbids discarding unique evidence merely to manufacture a documentation-size pass.

3 Risks and decisions

The improvement is real but smaller over the full solve than the early 4.31% cycle reduction: 407.006 seconds remains 1.357 times the engineering threshold and 128.819 seconds behind frozen Kissat. The paired full profiles use different complete trajectories, so broad percentages remain localization leads only. Any next repair still requires an exact common-state price plus exact candidate/order, scheduling, logical cost, and pruning comparison.

The repair is promoted because it is general, source-backed, trajectory-exact, proof-exact, and independently guarded. Completed semantic investigations are not reopened: no evidence contradicts Sweep parity, the directed bridge fix, BVE admission, or Vivify's normalized duplicate cleanup. Frozen Kissat is not rerun, and current instrumented Kissat remains diagnostic-only.

At user direction the asynchronous full Tier-D and surviving predecessor were canceled earlier to free capacity. Their frozen partial classifications remain non-scorable; resource and checker exits are not mislabeled as solver correctness failures, and both campaigns have zero pending certificate residue.

4 Next actions

Land this coherent increment and preserve its exact candidate pin. Then profile the repaired full Rank-27 trajectory against current instrumented Kissat and select the next earliest portable common-state owner where Tactix spends more for the same work or retains weaker state for comparable cost. Start with the remaining watched-propagation/search representation owners rather than changing behavior from cross-trajectory totals. Require exact ticks/costs, candidate and scheduling order, and resulting pruning before any behavioral repair. Any future promotion

repeats the complete frozen B28 plus fixed-C5 guard, checked models/proofs, zero solve loss, zero residue, and a checked `-c 2` Rank-27 witness under the 16-GiB/1,200-second contract.